ECON 101 End Term Exam

Question Paper

100 Marks (3 Hours)

Question 1 [20 Marks] – There are two individuals (A and B) who work in the same firm. Profits depend upon the joint efforts that these individuals decide to put into their work and all the profits are divided equally between the two individuals. Both individuals can either choose to devote high effort or low effort. If an individual decides to devote high effort then it costs that individual 50 while low effort costs only 10. If both of them decide to devote high effort then the firm earns a profit of 1000, while if both of them decide to devote low effort then the firm earns a profit equal to 500. If one individual decides to put in high effort and the other decides to put in low effort then the firm earns a profit of 950.

a) Write down the Normal Form Representation of this game. (5 Marks)

b) Are there any dominant strategies in this game, find the dominant strategy equilibrium? (5 Marks)

c) For each of the strategy combinations verify if that combination constitutes Nash equilibrium. (5 Marks)

d) If you think that a particular strategy combination in (c) constitutes a Nash Equilibrium, explain. (5 Marks)

Question 2 [20 Marks] – A firm produces output y using inputs x_1 and x_2. In order to do so it uses the following production technology: $y = 2x_1 + 3x_2$. Suppose that the price of input x_1 is 5 while the price of input x_2 is 10.

a) Assuming that the firm wants to produce y_0 level of output, state the cost minimization problem for the firm (5 Marks).

b) Use this information in order to solve for the cost minimizing level of inputs needed to produce y_0 units of good, describe the same using a well labelled diagram (5 Marks).

c) Solve for the cost function for this firm (10 marks).

Question 3 [20 Marks] – A firm has the following cost function: $C(Q) = Q^3 - 20Q^2 + 50Q + 100$. Use this information to answer the following –

a) Calculate the following – Marginal Cost, Average Cost, Fixed Cost, Average Variable Cost, Average Fixed Cost (5 Marks)

b) Use the above calculated values of average variable cost and marginal cost in order to describe the relationship between AVC and MC. Plot the above calculated AVC and MC on a well labelled diagram as well (7+3 Marks).

$\frac{2x_1}{3x_2} = \frac{1}{2} \times \frac{3}{3}$ $\quad 2\bar{x}_1 + 3\bar{x}_2 = y_0$ $\quad x_1 + 2x_2 = \bar{c}$

$\frac{4}{6} = \frac{3}{6} \Rightarrow 2\bar{x}_1 = y_0 - 3\bar{x}_2$ $\quad 3\bar{x}_2 = -2\bar{x}_1 + y_0$

$\Rightarrow \bar{x}_1 = \frac{y_0 - 3\bar{x}_2}{2}$ $\quad \Rightarrow \bar{x}_2 = \frac{2}{3}\bar{x}_1 + y_0$

475
2/9500
8
150
10

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c) The long run average cost curve lies above the various short run average cost curves. Do you agree with this statement? Explain. (5 Marks)

Question 4 [10 Marks] There are four individuals X, Y, A and B. X and Y **each** have one unit of an indivisible good, and no money. The value they attach to one unit of good is as follows - \$0 for X and \$3 for Y. A and B **each** have \$5 in money and no units of the good. The value they attach to one unit of good is as follows - \$5 for A and \$2 for B. No person has any value for a second unit of the good. The payoff of a person is defined as the value that person places on the unit of the good plus the money he/she has. Use this information in order to answer the following –

value + money.

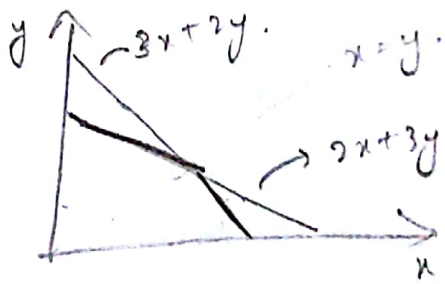
a) Explain whether the initial distribution of the good and money is pareto efficient or not. (5 Marks)

b) Suppose that X decided to sell her unit of good to B for \$2, Y sells her unit to A for \$3; explain whether this new distribution of good and money is a pareto efficient allocation or not. (5 Marks)

Question 5 [10 Marks] A monopolist producer faces a constant elasticity market demand curve, wherein the price elasticity of demand equals (-2). Also, the monopolist produces output at a constant marginal cost of 10 per unit. Use this information in order to calculate the price charged by the monopolist.

Question 6 [20 Marks] A consumer consumes two good x and y and the following utility function describes the preferences for the consumer - $U(x,y) = min\{2x + 3y, 3x + 2y\}$. The consumer has an income of 100, which is spent on consuming x and y. Price of good x is 5 while the price of good y is 10. Graph the indifference curves and budget line for the consumer. Use the same in order to solve for the chosen bundle and explain as well.

$\frac{3}{2}$, $\frac{2}{3}$



$2x + 3y = 3x + 2y.$

$\Rightarrow x = y.$

$2x + 3y < 3x + 2y.$

$\Rightarrow y < x$

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$U = x + y$ $\quad \frac{\partial U}{\partial x} = 3$ $\quad \frac{\partial U}{\partial y} = 2$ $\quad \frac{3}{2} = \frac{x}{y}$

$3x + 2y$ $\quad 120$

$\frac{dy}{dx} = \frac{3}{2}$ $\quad \frac{3 \times 3}{3 \times 3} = \frac{6}{9}$ $\quad \frac{120}{3} = 40, 60$

$\frac{3}{2}$

$\frac{3}{2}$ $\quad 120$

Final Examination
ECO 101: Principles of Microeconomics
Department of Economics
Shiv Nadar University
Monsoon 2017.

This is a two and a half hour long closed book, closed notes in class examination. Answer all the questions, and be clear and precise in your answers. Any act of *academic dishonesty* will be taken very seriously. *All the best.*

$3x + 2y$

Question 1: Consumer Choice (30 points)

Consider an individual, Ryan, with the following a preference over X and Y that is represented by $U(x, y) = 3\min\{x, y\} + 2\max\{x, y\}$. Ryan has 120 INR at his disposal to spend over X and Y, and the prices of good X and good Y are given by 3 and 1 respectively.

$y > x$

(a.) Draw the family of indifference curves for Ryan and explain the shape. (10 points)
(b.) Solve for the optimal consumption bundle for Ryan in this situation. Draw this optimal choice in a properly labelled diagram. (10 points)
(c.) Now suppose the price of good Y has increased to 2, keeping that of good X at the previous level. How does change the optimal choice of Ryan? Show that in a diagram. Without doing any further calculation, just from the diagram can you comment how the demand for good X looks like? (10 points)

Question 2: Cost curves and firm supply function (30 points)

Consider a firm in a competitive industry with a cost function $C(q) = aq^3 + bq^2 + cq + d$, where $a > 0$, $b < 0$, $c > 0$ and $d > 0$, where $q > 0$ is the output the firm produces.

(a.) Derive the expressions for Total Variable Cost, Total Fixed Cost, Average Variable Cost, Average Fixed Cost and the Marginal cost. (10 points)
(b.) Draw the aforementioned curves in a clearly labelled diagram, clearly pointing out the curvature and critical points and carefully explain the interrelationship between the average and marginal schedules. (10 points)
(c.) Derive the firm short run supply curve, and show this in a diagram. Briefly mention the shut down decision that was critical in the derivation of the supply curve. (10 points)

Question 3: Simultaneous move game (40 points)

Imagine two big oil companies-Indian Oil (IO) and Bharat Petroleum (BP)-own adjacent oil fields. Under the fields is a common pool of oil worth INR 12 million. An oil company needs to drill a well to extract oil from underneath the fields. Suppose each company has a choice to drill either one or two wells. Drilling a well to recover the oil costs INR 1 million. If each company drills one well, each will get half of the oil. If one of them drills two wells when the other has only one, that company (with two of the three wells) gets two-thirds of the oil and the other one gets one-third of the oil. Finally, if each company drills two

$= \frac{1}{x_2}$

$3 \times 2 = 2 + 6$ $\quad 20$

$3 \times 2 = 2 \times 1$

$TVC = TFC$

$\Rightarrow \frac{d}{q} = d$

$\Rightarrow q = 1$

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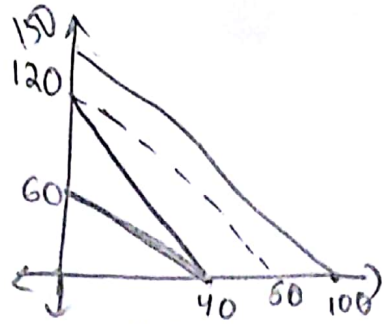


$3x + 2y = 120$

$(0, 60)$

well, they again split the oil reserve equally. Each company will however have to pay for as many wells as it drills.

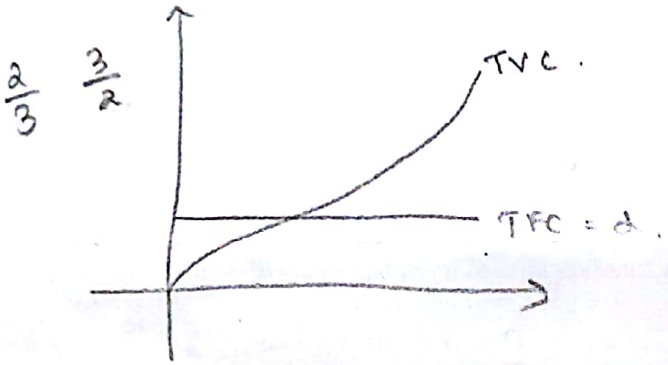
(a) Formulate the situation as a normal form game. (8 points)

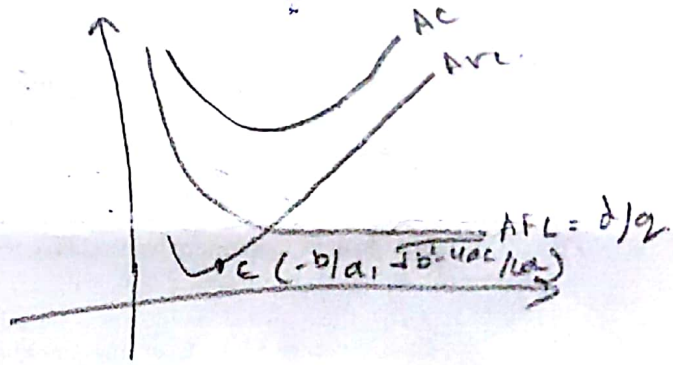
(b) Does any of the player's has a dominant strategy? If yes, which one of the strategies survive the iterated elimination of strictly dominated strategies? (10 points)

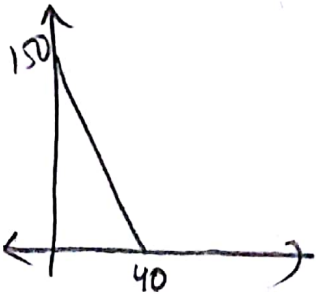
(c) Find out the Nash equilibrium of the game. Briefly explain the mechanism of finding the Nash equilibrium. Does the solution coincide with the answer of part (b)? Provide intuition behind your answer. (15 points)

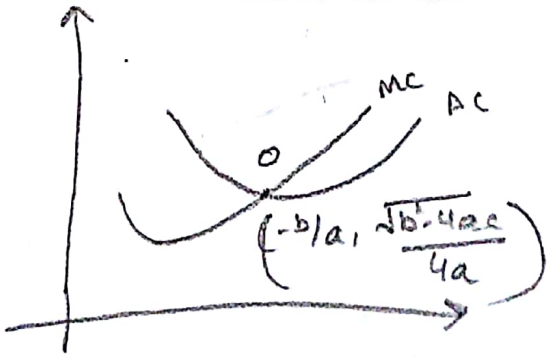
(d) Briefly summarize impact of the equilibrium that emerges in this game on the society. Can you provide an example where such a strategic behaviour is commonly observed? (7 points)

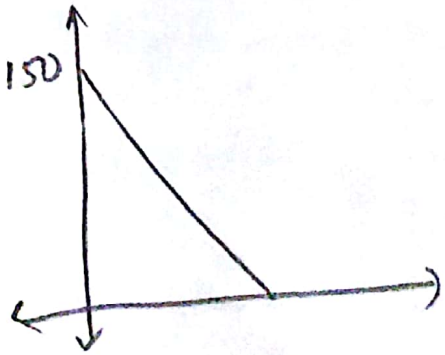
Date: December 4, 2017.
Instructor: *Subhra K. Bhattacharya.*

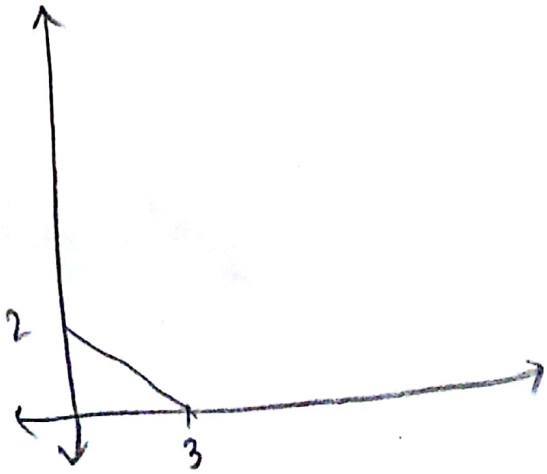












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